

12 Numerical Simulation Methods and Their Applications

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12.1 Overview

Computer simulations are mathematical modelling techniques that are widely applied in studying various scientific, engineering, economic and social science subjects. These algorithms are based on the principles of classical and/or statistical mechanics, probability and statistics, numerical solutions of ordinary and stochastic differential equations, statistical sampling methods, etc. Two classes of simulation techniques, namely Monte Carlo (MC) and molecular dynamics (MD) simulation methods, have been widely used in studying particle-based systems, such simple liquids, polymers and biomolecules. In this project, students can choose to work on one of the MC or MD simulation methods and its applications in studying the structural and/or dynamic properties of one representative particle-based system.

It will be helpful but not necessary if the students have some programming experience in using any language, such as Matlab, Python, C/C++ etc.

12.2 Link to global sustainability

The simulation methods developed in this project can be applied to study the structural, dynamical and mechanical properties of various soft matter systems, such as associative polymers, biopolymers (proteins, peptides, and DNA) and bio-mimicking materials. The simulation results and the insights gained from them can provide guidance for designing novel functional materials which are recyclable, reprocessible and environment-friendly, as well as highly efficient antimicrobial and antiviral drugs, and so contribute to global sustainability.

12.3 Pre-requisites

Students may find materials from one or more of the modules MA2NAN, MA2MOD, MA2MPH, MA3MP and MA3APS to be useful for this project, but it is not a requirement that students have taken them.

12.4 Questions

Students can choose to work on one of the following topics:

- Monte Carlo Methods and Applications in Particle-based Systems (e.g., Metropolis method for simulating Lennard-Jones liquid or single bead-spring chain);
- Molecular Dynamics and Finite Difference Methods (i. e., finite difference methods for numerical solutions of Newton's equation of motion, MD simulation of Lennard-Jones liquid or single bead-spring chain);
- Molecular Dynamics Methods in Different Ensembles (e.g., numerical solutions of Newton's equation of motion, MD simulations of Lennard-Jone liquids in microcanonical (NVE) and canonical (NVT) ensembles)

In each project, students are expected to address the following questions:

Question 1 [5 marks]

Write a brief review about the development of computer simulation methods, starting from the Monte Carlo method developed in the Manhattan Project to later development of MC and MD methods with the enhanced computational power.

Question 2 [10 marks]

For students working on MC methods, i) describe one or two pseudo random number generators (PRNG) and give examples to show how the PRNG(s) work; ii) describe the simple and importance sampling methods and, if having time, give examples on using these sampling methods for numerical integrations, such as calculating the value of π ; iii) discuss the statistical mechanics background of Monte Carlo methods; iv) derive the Metropolis method by using the concepts of importance sampling and detailed balance.

For students working on MD methods: i) discuss the classical mechanics background of equations of motion; ii) a) If working on MD with finite difference methods, derive two or more finite difference methods for numerical solutions of Newton's equation of motion (2nd order ODE) using Taylor expansion; b) If working on MD in different ensembles, derive at least one finite difference method and discuss at least one thermostat for maintaining constant temperature in the canonical ensemble; iii) discuss the orders of accuracy in calculating the positions and velocities of the particles by using different finite difference methods.

Question 3 [2 marks]

Describe and sketch the Lennard-Jones potential and its application in modelling simple liquids.

Describe the concept of periodic boundary conditions (PBC) as used in simulating particle-based systems and insert a figure to demonstrate how PBC works.

Question 4 [10 marks]

Describe the steps for implementing the selected MC or MD methods into a simulation code for studying simple liquid or single bead-spring chain, including initial configuration generation, loops for generating the moving trajectories of the particles and post-simulation data analysis, etc.

Develop your own simulation code using any programming language OR download and describe a simulation code found in open resources. You must clearly state the origin of the simulation code, either self-developed or downloaded from open resource (give the link or citation of the resource).

Question 5 [20 marks]

Apply the developed or downloaded code to simulate a simple Lennard-Jones liquid with 50-200 particles or a single bead-spring chain with 20-50 beads and demonstrate the equilibration process of the system by calculating the total energy of the system over the simulation steps and showing the results in plot(s).

Perform numerical analysis of some structural and dynamical properties of the system, such as the radial distribution function of particles in the simple LJ liquids or the end-to-end distance and radius of gyration of the single bead-spring chain, the mean-squared displacements and diffusion coefficients of particles in the system, etc.

If not running a simulation code, find and discuss simulation results on the structural properties of LJ liquids or bead-spring polymer chains in literature, and derive the mathematical formulae for predicting these structural and dynamical properties. You must clearly state the resource of these results.

Question 6 [3 marks]

Draw conclusions and give perspectives of the study.

Question 7 [5 marks] For MA3PPRNR students (NUIST students who are studying in Reading),

Write a literature review about theoretical calculations of the conformational and/or dynamical properties of ideal polymer chains (e. g., see Ref. 4 below). This review can be integrated with your answer to Question 5.

12.5 References

1. Computer Simulation of Liquids, M. P. Allen and D. J. Tildesley, Oxford Science Publications, 1987;
2. Understanding Molecular Simulations: From Algorithms to Applications, Daan Frenkel and Berend Smit, Academic Press, 2001;
3. The Art of Molecular Dynamics Simulation, D. C. Rapport, Cambridge University Press, 2004.
4. Polymer Physics, Michael Rubinstein and Ralph H. Colby, Oxford University Press, 2003.
5. Any other books or online resources about computer simulations.